

Question Paper

Exam Date & Time: 28-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

VLSI Testing and Testability [ECE_3125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

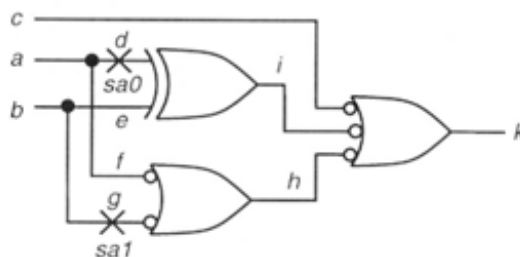
- 1) A mixed-signal VLSI chip has 1,024 digital pins. To reduce the testing cost, it is decided to design the digital part with built-in self-test (BIST) circuitry. Thus, most of the digital signals can be generated or analyzed on the chip and do not have to be supplied from/to the ATE. It can now be tested on a 256-pin ATE. Calculate the per second cost of testing. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) An electronics manufacturing plant receives an order for supplying a large number of single-board systems. Components for the boards are purchased and stored in a warehouse whose rent is \$10,000 per day. One technician assembles and tests one system each day. The cost of workspace (building, equipment, etc.) is estimated as \$500n2 per day for n technicians. (5)
- (a) How many technicians should the plant employ on the job?
- (b) If the cost of components for a system is \$10,000 and the daily wages of a technician are \$200, then what is the minimum cost of one system?

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) (i) Derive the fault indistinguishability condition (5)
- A) $f_1(V) \oplus f_2(V) = 0$ from $[f_0(V) \oplus f_1(V)] \oplus [f_0(V) \oplus f_2(V)] = 0$.
- (ii) For the below circuit, show that the two faults d s-a-0 and g s-a-1 are equivalent.



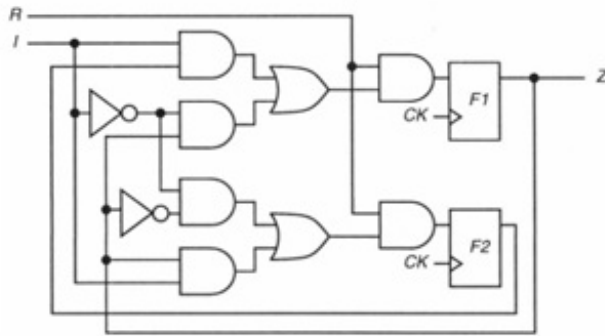
Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Determine the number of faulty sites, number of faults, and minimum test length for 100% single-stuck-at-fault coverage with the truth table for the following gates: (5)
- (i) NAND (ii) NOR

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- (ii) Need for partitioning
- (iii) Partitioning at different levels
- (iv) Five golden rules of partitioning.

B) Draw the s-graph for the circuit in Figure below. Identify the flip-flop to be scanned such that all cycles are broken.



Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) What is the requirement for Built-in-Self Test (BIST). Explain its hierarchy, implementation, and pattern generation in detail. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- B) Design the pseudo-exhaustive pattern generation for 8 vector testing and 20 vector testing with proper truth tables using the MUX implementations. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECE_3125:VLSI Testing and Testability

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECE_3125:VLSI Testing and Testability

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)